Cross-language numerical comparison

Repository source: expected/CROSS_LANGUAGE_COMPARISON.md

The Python stress test was executed in the assistant's container. The matching base-R stress test was executed by Bruce J. Swihart in his local R environment. The two programs are independent implementations and use language-specific random-number generators, so they do not test the same random tensors even though both runs are deterministic within their own language.

Check	Python	R	Interpretation
Maximum pointwise determinant discrepancy	`1.819e-12`	`2.046e-12`	Direct projected-matrix determinant and compact tensor-contraction formula agree to floating-point precision.
Maximum relative error in `integral |A_Y|^2`	`2.829e-15`	`2.038e-15`	Numerical quadrature agrees with `(176*pi/7)*tau`.
Maximum relative error in `integral (4 det A_Y)^2`	`3.427e-15`	`5.107e-15`	Numerical quadrature agrees with the proposed quartic invariant identity.
Candidate volume coefficient	`0.383602704709067716`	`0.383602704709068`	Agreement to all 15 digits printed by R.

All programmed numerical-stress acceptance thresholds passed in both languages. The finite pairing-count table also passed independent Python and base-R enumerators. The repository's first hosted GitHub Actions run subsequently completed all three verification jobs successfully.

This is strong cross-language numerical evidence for the tensor conventions and constants. It is not a proof of the full lower bound. The general symbolic integration is implemented in Python/SymPy, while the full mathematical argument is presented in the LaTeX research draft and still awaits independent subject-matter review.